La Scuola d'Italia / Dr. Huson / IB Math
24 April 2026

First & Last name:

7.2 Classwork: Log Challenge (no calculator)

1. [Maximum mark: 7]

   Solve $\log_2(x-3) = \log_4(x+9)$.

2. [Maximum mark: 15]

Consider the series $\ln x + p\ln x + \frac{1}{3}\ln x + \ldots$, where $x \in \mathbb{R}$, $x > 1$ and $p \in \mathbb{R}$, $p \neq 0$.

(a) Consider the case where the series is geometric.

(i) Show that $p = \pm\frac{1}{\sqrt{3}}$.

(ii) Given that $p > 0$ and $S_\infty = 3 + \sqrt{3}$, find the value of x. [5]

(b) Now consider the case where the series is arithmetic with common difference d.

(i) Show that $p = \frac{2}{3}$.

(ii) Write down d in the form $k\ln x$, where $k \in \mathbb{R}$.

(iii) The sum of the first n terms of the series is $-3\ln x$. Find the value of n. [10]